Modul 11 Tahap 1

1. Identifikasi nama, versi, dan storage engine

```sql
-- Modul 11 - Tahap 1: Eksplorasi Perkembangan dan Kelengkapan DBMS
-- Proyek: Green Map PNM
-- Prasyarat: database db_green_map_pnm sudah dibuat (sql/01_schema_data.sql)

USE db_green_map_pnm;

-- 1. Identifikasi DBMS
-- Mengetahui versi DBMS
SELECT VERSION();

-- Mengetahui informasi DBMS
SELECT @@version_comment;

-- Mengetahui database yang tersedia
SHOW DATABASES;

-- Melihat storage engine
SHOW ENGINES;

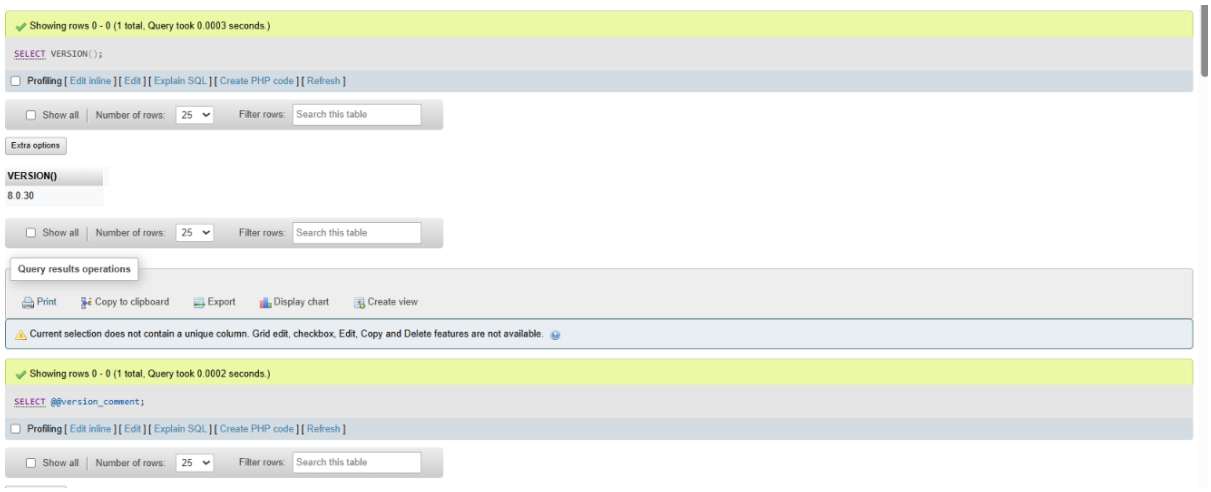
-- Melihat karakteristik server
SHOW VARIABLES LIKE 'character_set_server';

-- Melihat collation
SHOW VARIABLES LIKE 'collation_server';

-- Memeriksa informasi waktu server
SELECT NOW();

-- Memeriksa database aktif
SELECT DATABASE();
```

Bukti Screenshot



@@version_comment
MySQL Community Server - GPL

Show all | Number of rows: 25 Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

```
SHOW DATABASES;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Database
classicmodels
db_green_map_pnm
db_kepegawaian
db_modul_9
etle_db
information_schema
mysql
performance_schema
sdg_transport_db
sdg_transport_db_recovery
sys
trpl_admin

Query results operations

Print Copy to clipboard Create view

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

```
SHOW ENGINES;
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Engine	Support	Comment	Transactions	XA	Savepoints
MEMORY	YES	Hash based, stored in memory, useful for temporary...	NO	NO	NO
MRG_MYISAM	YES	Collection of identical MyISAM tables	NO	NO	NO
CSV	YES	CSV storage engine	NO	NO	NO
FEDERATED	NO	Federated MySQL storage engine	*NULL*	*NULL*	*NULL*
PERFORMANCE_SCHEMA	YES	Performance Schema	NO	NO	NO
MyISAM	YES	MyISAM storage engine	NO	NO	NO
InnoDB	DEFAULT	Supports transactions, row-level locking, and fore...	YES	YES	YES
BLACKHOLE	YES	/dev/null storage engine (anything you write to it...	NO	NO	NO
ARCHIVE	YES	Archive storage engine	NO	NO	NO

Query results operations

Print Copy to clipboard Create view

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Query results operations

Print Copy to clipboard Create view

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

```
SHOW VARIABLES LIKE 'character_set_server';
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

Variable_name	Value
character_set_server	utf8mb4

Query results operations

Print Copy to clipboard Create view

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Your SQL query has been executed successfully.

```
SHOW VARIABLES LIKE 'collation_server';
```

Profiling [Edit inline] [Edit] [Create PHP code] [Refresh]

Extra options

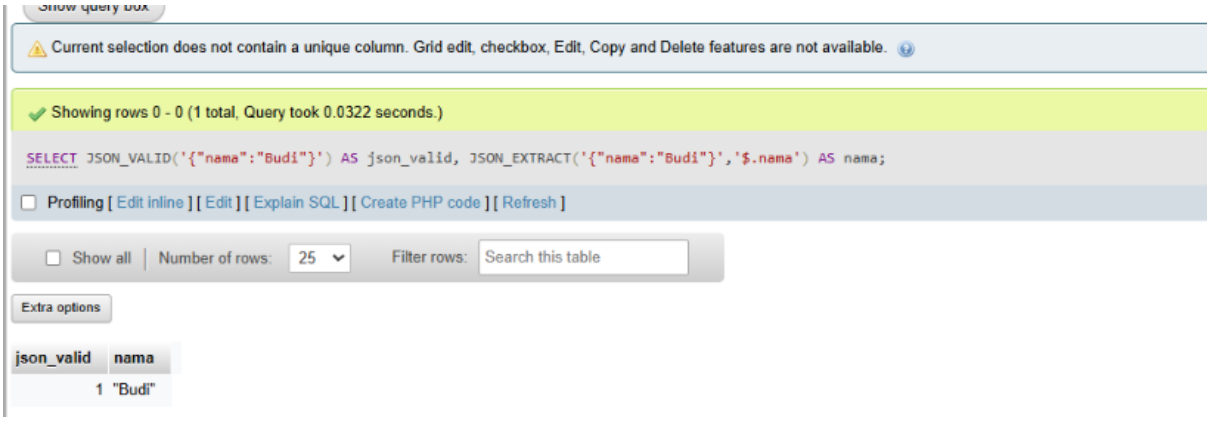
Variable_name	Value
collation_server	utf8mb4_0900_ai_ci

2. Fitur JSON

- **Query 1**

```sql
-- 2. Fitur JSON
-- Memeriksa dukungan fungsi JSON
SELECT
    JSON_VALID('{"nama":"Budi"}') AS json_valid,
    JSON_EXTRACT(
        '{"nama":"Budi"}',
        '$.nama'
    ) AS nama;
```

Bukti Screenshot



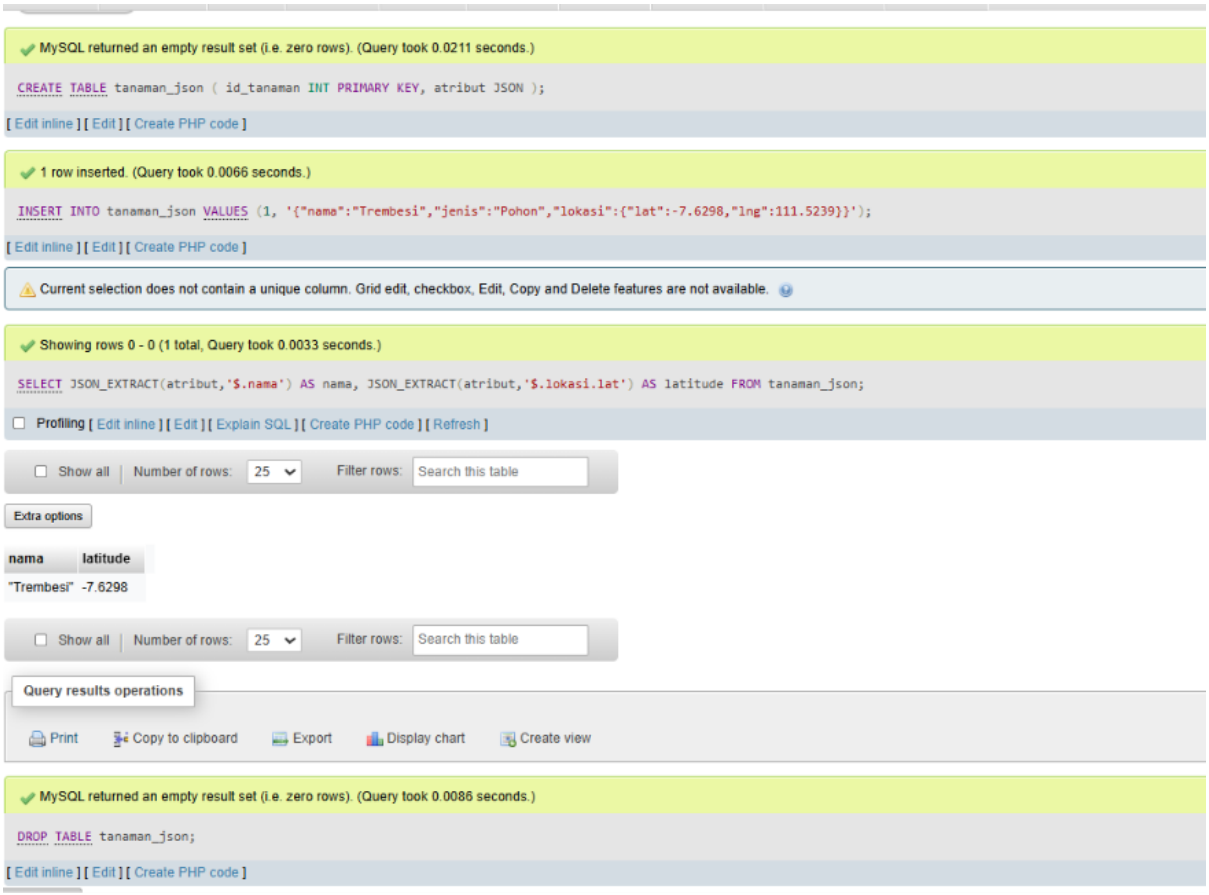
- **Query 2**

```sql
-- Membuat tabel percobaan dengan kolom JSON
CREATE TABLE tanaman_json (
  id_tanaman INT PRIMARY KEY,
  atribut JSON
);

-- Mengisi data atribut tanaman dalam format JSON
INSERT INTO tanaman_json VALUES
(1, '{"nama":"Trembesi","jenis":"Pohon","lokasi":{"lat":-7.6298,"lng":111.5239}}');

-- Mengambil nilai nama dan latitude dari JSON
SELECT JSON_EXTRACT(atribut,'$.nama') AS nama,
       JSON_EXTRACT(atribut,'$.lokasi.lat') AS latitude
FROM tanaman_json;
```

Bukti Screenshot

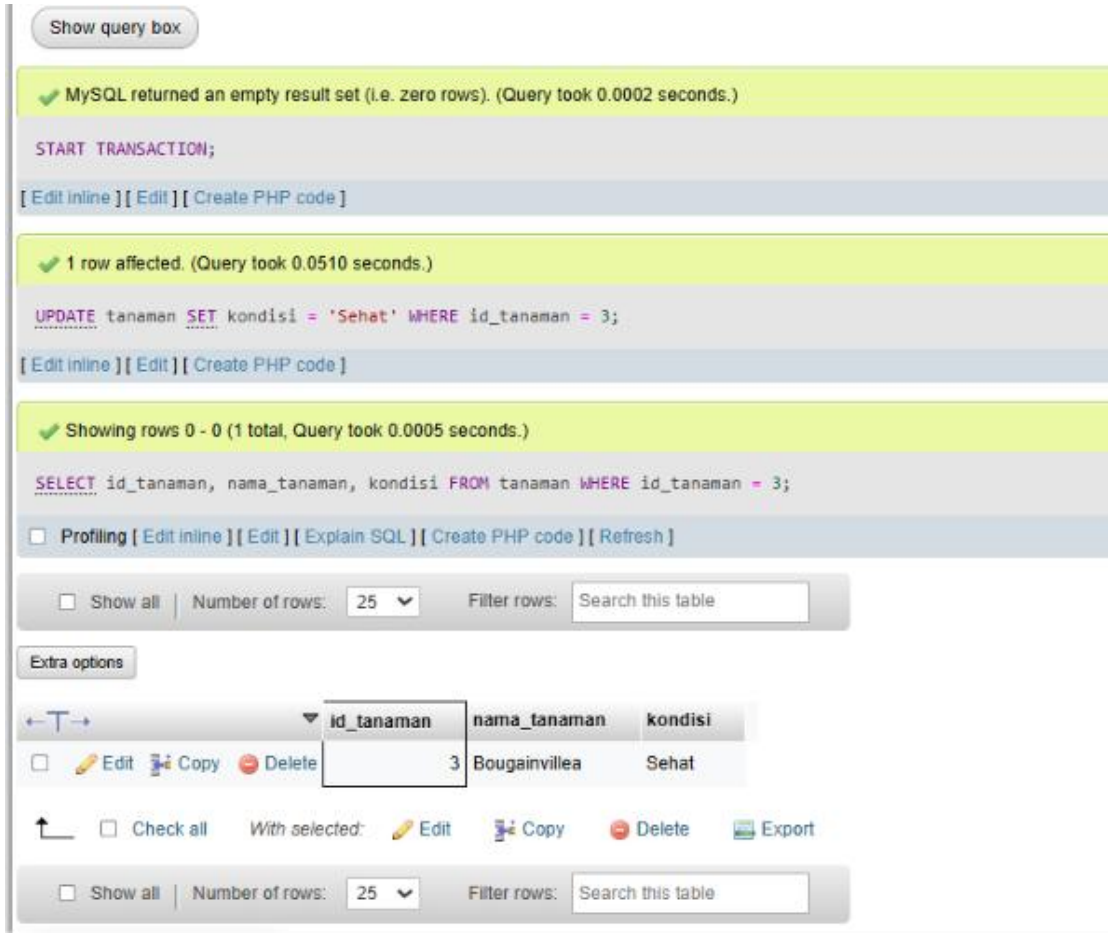


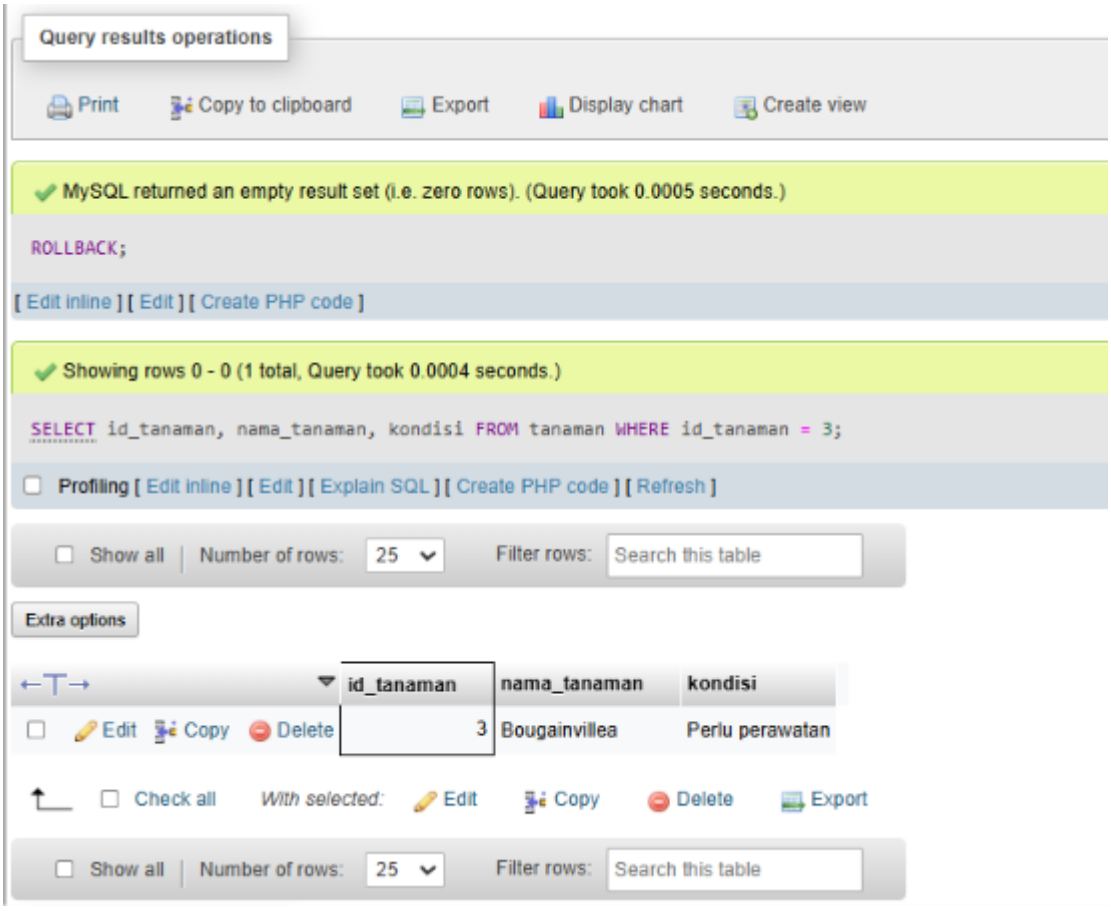
3. Fasilitas Transaksi

- **Query 3**

```sql
-- 3. Fasilitas transaksi (data dibatalkan dengan ROLLBACK)
-- Memulai transaksi
START TRANSACTION;
-- Mengubah kondisi tanaman sementara
UPDATE tanaman SET kondisi = 'Sehat' WHERE id_tanaman = 3;
SELECT id_tanaman, nama_tanaman, kondisi FROM tanaman WHERE id_tanaman = 3;
-- Membatalkan perubahan
ROLLBACK;
SELECT id_tanaman, nama_tanaman, kondisi FROM tanaman WHERE id_tanaman = 3;
```

Bukti Screenshot





4. Hak akses, fungsi, dan log biner

- **Query 4**

```sql
-- 4. Hak akses pengguna
-- Membuat pengguna baru
CREATE USER 'pengguna_baca'@'localhost' IDENTIFIED BY 'Baca#2026';
-- Memberi hak baca saja
GRANT SELECT ON db_green_map_pnm.* TO 'pengguna_baca'@'localhost';
-- Melihat hak akses pengguna
SHOW GRANTS FOR 'pengguna_baca'@'localhost';
```

Bukti Screenshot

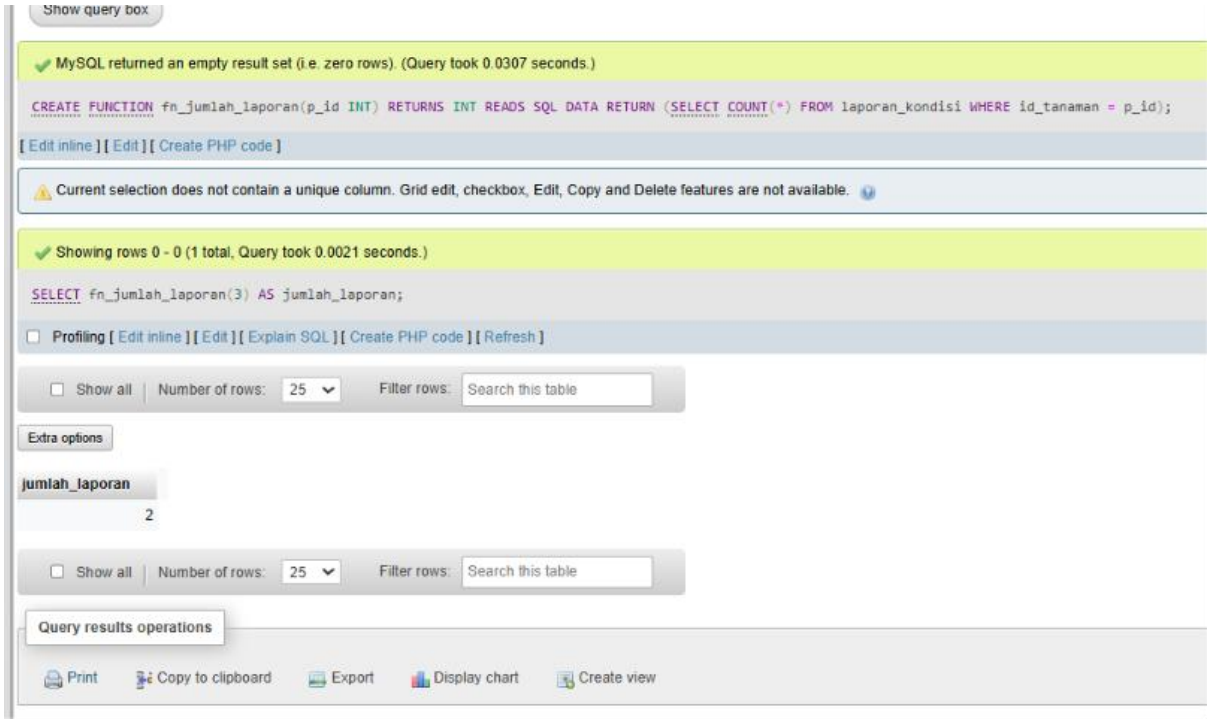


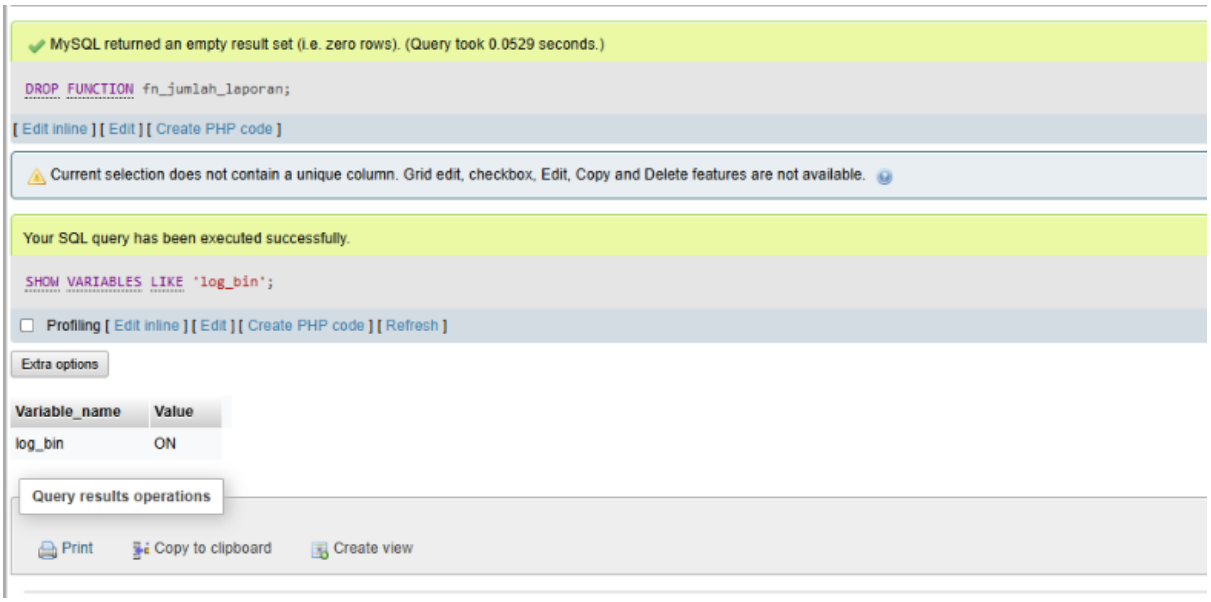
- **Query 5**

```sql
-- 5. Stored function dan log biner
-- Membuat fungsi penghitung laporan per tanaman
CREATE FUNCTION fn_jumlah_laporan(p_id INT) RETURNS INT
READS SQL DATA
RETURN (SELECT COUNT(*) FROM laporan_kondisi WHERE id_tanaman = p_id);

-- Memanggil fungsi untuk tanaman dengan id 3 (Bougainvillea)
SELECT fn_jumlah_laporan(3) AS jumlah_laporan;
-- Memeriksa log biner
SHOW VARIABLES LIKE 'log_bin';
```

Bukti Screenshot





5. Ringkasan hasil identifikasi

Tugas	Hasil
Nama dan versi DBMS	[isi dari Query 1]

Storage engine	InnoDB (default), mendukung transaksi dan Foreign Key
Fitur JSON	Tersedia: tipe JSON, JSON_VALID, dan JSON_EXTRACT berhasil dijalankan
Fasilitas transaksi	Tersedia: START TRANSACTION, COMMIT, dan ROLLBACK berhasil dibuktikan
Fitur lain	Pengaturan hak akses (GRANT, REVOKE), stored function, dan log biner

Fitur yang paling relevan untuk proyek

Menurut kami fitur yang paling relevan untuk Green Map PNM adalah integritas data lewat Foreign Key dan transaksi pada InnoDB. Alasannya, laporan kondisi dan log perawatan selalu terkait dengan satu tanaman dan satu pengguna, sehingga data tidak boleh menggantung tanpa induknya, dan perubahan data harus tersimpan utuh (KNF-02). Fitur berikutnya adalah hak akses pengguna karena SKPL membatasi pengelolaan data hanya untuk pengelola (KNF-08), serta indeks agar pencarian tanaman tetap cepat ketika data bertambah (KNF-06). Fitur JSON bersifat tambahan, berguna bila nanti atribut tanaman makin beragam.